01 / Renal physiology: scope and study route

Lecture PDF pp. 1, 15–16, 47 | Correct lecture: BBMS2006 (Renal - student version) (1).pdf, Dr J Chu

Follow three connected pathways

Juxtamedullary loop → medullary osmotic gradient → collecting-duct water / urea handling.
Body-fluid change → brain sensing → VP, thirst and RAAS responses.
Hormone deficiency / excess → altered water, sodium, potassium and clinical signs.

Lecture learning target	Notes pages
Nephron types; countercurrent exchange versus multiplication; medullary gradient and vasa recta.	2–6
Dilute / concentrated urine; AQP2, AQP3/4; urea, UT-A1 / UT-A3; classroom hypertonic-saline question.	7–10
Body-fluid distribution and ions; buffering; VP synthesis, storage and release.	11–15
Osmotic / haemodynamic sensing; baroreflex; classroom cardiovascular / occlusion questions.	16–20
V_2 / V_1a effects; thirst; all four ANGII actions; aldosterone transporters and integrated circuits.	21–28
DI, SIADH, exercise-related hyponatraemia, aldosterone disorders and electrolyte manifestations.	29–34

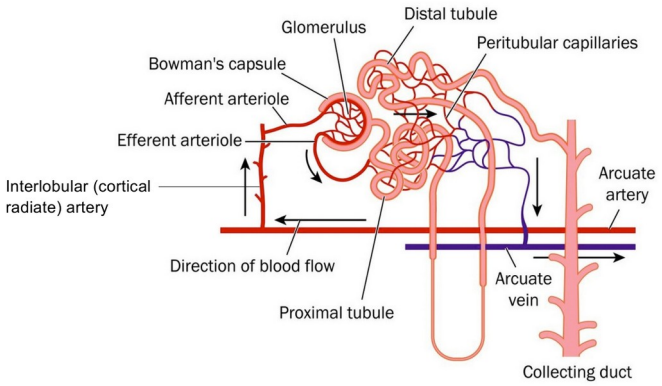
Classroom material remains beside its mechanism

Original photograph	Placement
Photos 1–2: duplicate views of pressure / VP supplement.	19: baroreflex and cardiovascular effects.
Photo 3: Activity 3, small hypertonic-saline injection.	9: VP, AQP2 and UT-A1.
Photo 4: Activity 4 plus vagotomy follow-up.	20: opposing carotid and aortic signals.
Photo 5: Activity 6, water restriction and urine concentration.	29: diabetes insipidus.
Photo 6: Lucas's hot half-marathon and water intake.	31: dilutional hyponatraemia.

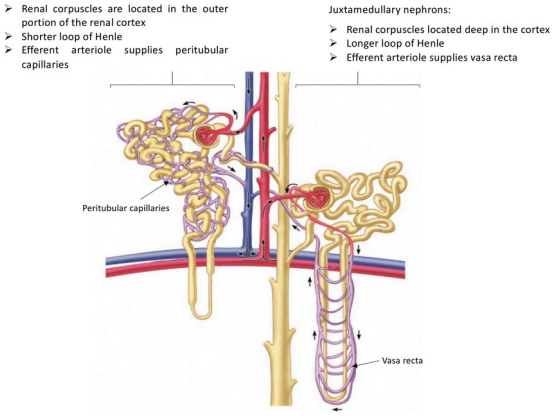
This document uses the newly supplied RENAL lecture, not Urinary System 2026.pdf. All references are actual PDF pages. Classroom photos are transcribed; teaching figures from the PDF retain their labels and annotations. [Extension] identifies supporting physiology; source wording needing correction is explicitly quoted / qualified. Two integrated self-written questions are included; they are not predictions of actual exam content. Coverage checks are on pp. 35–37.

02 / Nephron structure and the two capillary beds

Lecture PDF pp. 2–3



Original lecture figure, PDF p. 2.



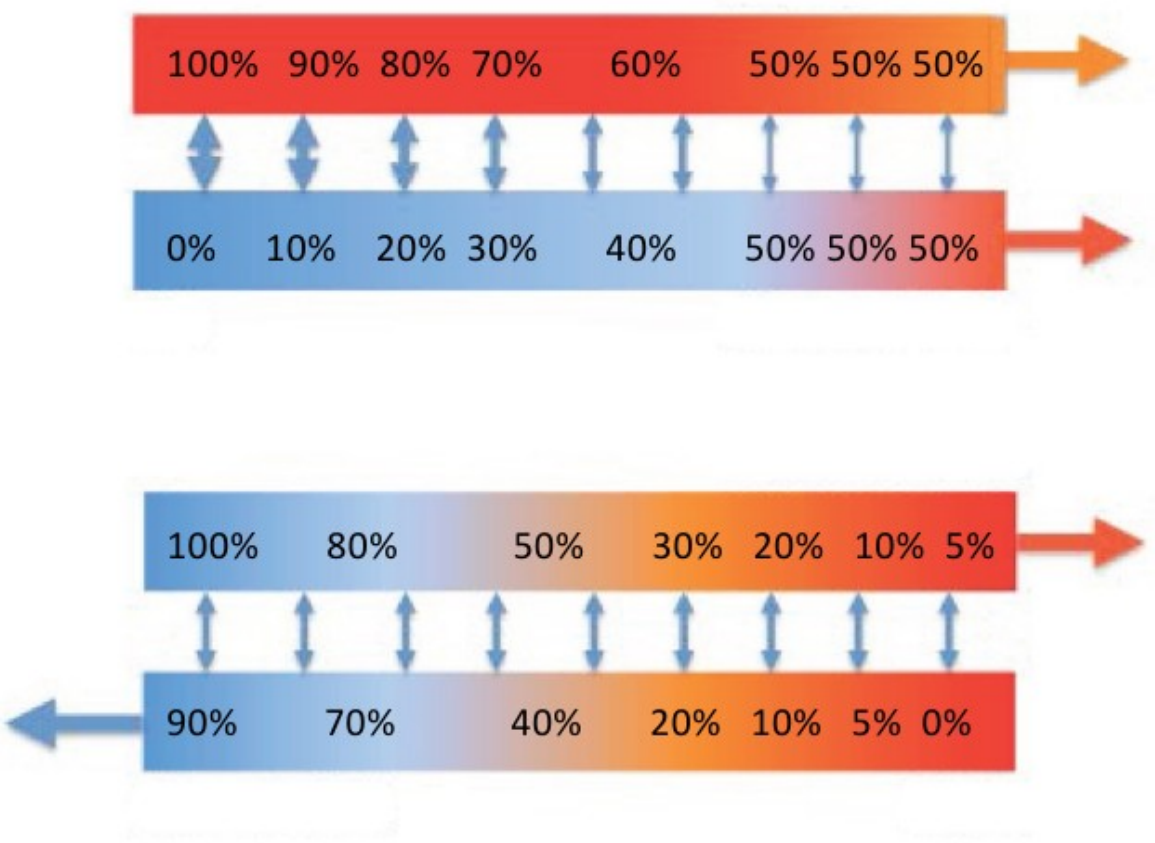
Original lecture figure, PDF p. 3.

Key written point	Detail
Functional units	Each kidney contains around ONE MILLION nephrons, held together by connective tissue.
Nephron components	Renal corpuscle = glomerulus + Bowman’s capsule. Nephron = renal corpuscle + renal tubule.
Two capillary beds in series	Afferent arteriole → glomerulus → efferent arteriole → peritubular capillaries / vasa recta.
Tubular labels	Proximal tubule → loop of Henle → distal tubule → collecting duct. Collecting ducts receive fluid from multiple nephrons.
Vascular labels	Interlobular / cortical radiate artery; arcuate artery and vein; afferent / efferent arterioles; direction of blood flow.

Compare	Cortical nephron	Juxtamedullary nephron
Corpuscle position	Outer renal cortex.	Deep cortex, near corticomedullary boundary.
Loop of Henle	Shorter.	Longer, extending deeply into medulla.
Efferent supply	Peritubular capillaries.	Vasa recta.
Concentration role	Less important for building the deep gradient.	Long-loop countercurrent multiplication is central to the concentrating gradient.

03 / Co-current versus countercurrent exchange

Lecture PDF p. 4



Original lecture image, including visible teaching annotations (PDF p. 4).

Feature	Co-current	Countercurrent
Flow directions	Same direction.	Opposite directions.
Exchange along the length	Driving difference diminishes; both streams approach the same composition.	A local difference can be maintained along successive positions.
Source example	100% versus 0% approaches 50% / 50%.	Source outlet example approaches 5% in one stream and 90% in the other.
Amount transferred	Limited by early equilibration.	Higher maximum transfer under comparable conditions.

Retain the definition

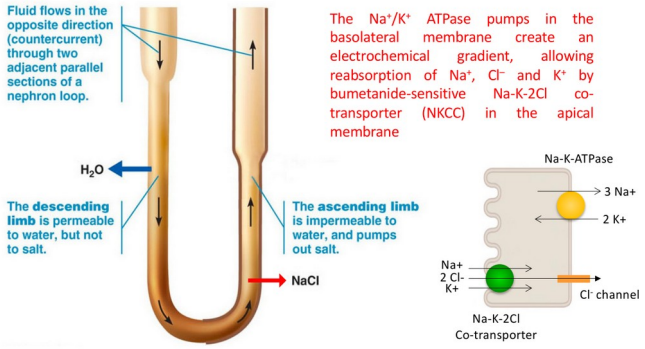
Countercurrent EXCHANGE is passive equilibration of material between two streams moving in opposite directions.

The source says the systems “do not reach equilibrium”: read this as preservation of a local driving difference during flow, not a claim that thermodynamics forbids equilibrium.

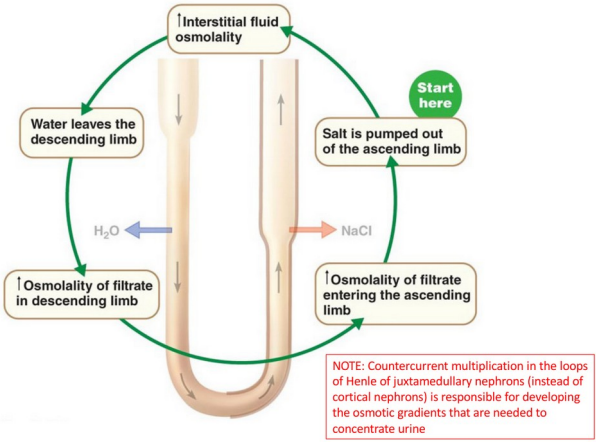
Exchange preserves / transfers an existing concentration difference; multiplication uses energy plus differential permeability and repeated flow to build an axial gradient. Do not use these two terms interchangeably.

04 / Countercurrent multiplication: single effect

Lecture PDF pp. 5–6



Original lecture figure, PDF p. 5.



Original lecture figure, PDF p. 6.

Written / figure detail	Mechanism
Energy requirement	Countercurrent multiplication uses energy to generate a gradient that later enables tubular water reabsorption and concentrated urine.
Descending limb	Highly permeable to water; relatively low salt permeability in the simplified slide model.
Ascending limb	Low water permeability; the thick ascending limb reabsorbs salt while leaving water behind.
Apical transporter	Bumetanide-sensitive Na^+–K^+–$2Cl^-$ cotransporter, NKCC2, moves these ions from lumen into thick-ascending-limb cells.
Basolateral energy source	Na^+/K^+-ATPase moves 3 Na^+ out / 2 K^+ in → electrochemical gradient supports NKCC2. Cl^- exits through basolateral channels.
Essential nephron type	Long loops of JUXTAMEDULLARY nephrons generate the deep gradient, rather than short cortical loops.

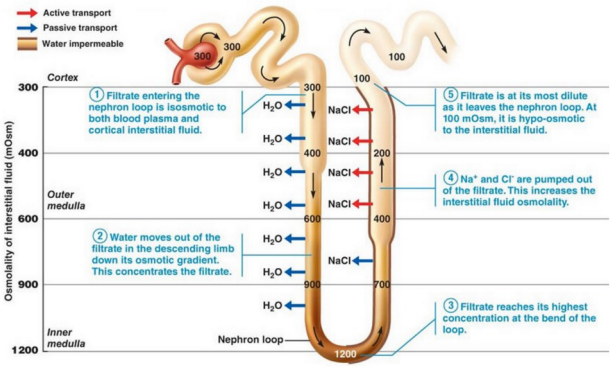
Repeat the cycle

Ascending salt reabsorption → interstitial osmolality ↑
→ descending water loss → descending tubular fluid concentrates
→ concentrated fluid reaches ascending limb → more salt available to remove.
Continuous flow repeats the single effect along the loop.

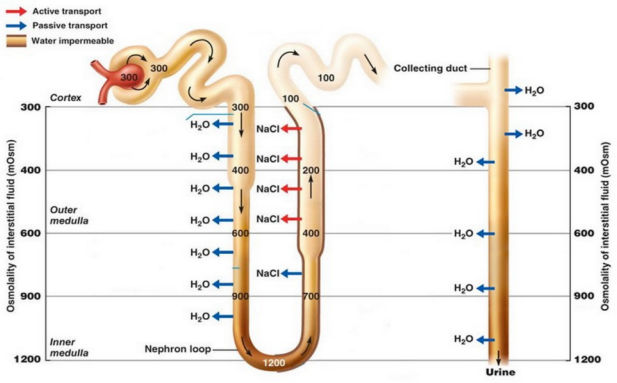
NKCC is a cotransporter, not a channel, despite “NKCC channels” in the source summary. The ATPase supplies energy indirectly: NKCC2 is secondary active transport. The thick ascending limb is the principal active salt-removing segment; not every ascending segment pumps salt in the same way.

05 / Why the loop concentrates, then dilutes filtrate

Lecture PDF pp. 7–8



Original lecture figure, PDF p. 7.



Original lecture figure, PDF p. 8.

Source sequence	What to follow
1. Entering loop	Filtrate initially roughly isotonic with blood / cortex, around 300 mOsm in the diagram.
2. Descending limb	Water leaves toward the increasingly hyperosmotic interstitium → tubular fluid concentrates.
3. Loop tip	Highest concentration, up to the illustrated 1200 near the deepest tip.
4. Ascending limb	NaCl leaves while little water follows → tubular fluid becomes progressively dilute.
5. Leaving loop	Illustrated minimum around 100 mOsm; fluid is hypo-osmotic to interstitium.

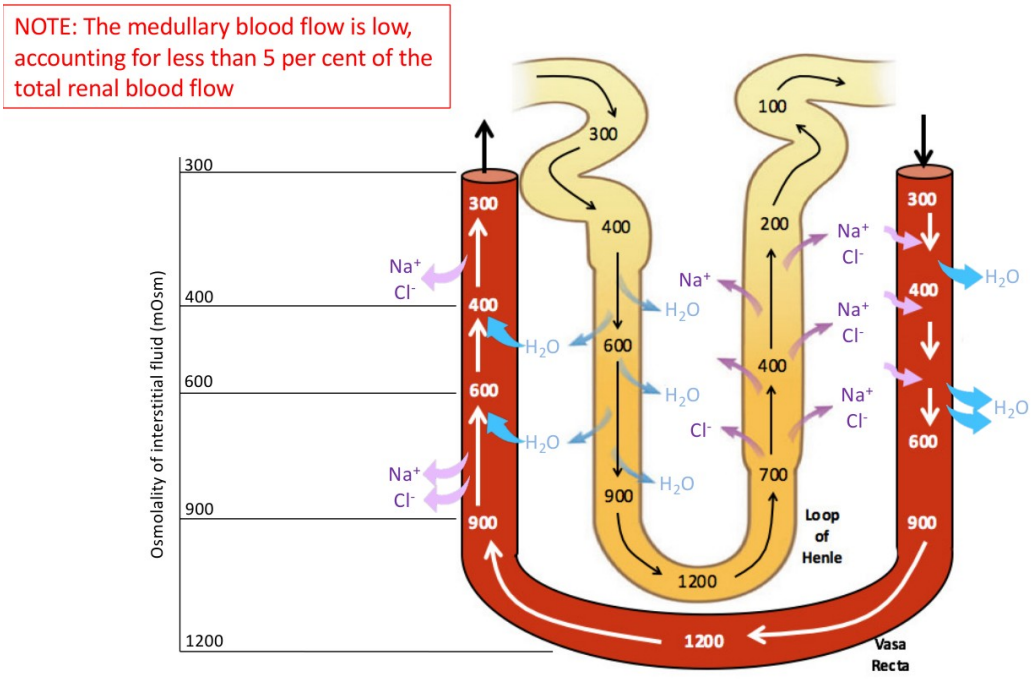
Read the longitudinal gradient, not just one cross-section

Cortex / corticomedullary region ~300 → outer-medullary levels ~400 and 600 → deeper medulla ~900 → papillary tip ~1200 in the teaching figure.
Longer loop → larger potential corticomedullary osmotic gradient.

The figures distinguish active versus passive transport and water-permeable regions. These values illustrate a concentrating kidney and are not fixed at every hydration state. The collecting duct passes down the same gradient; its water permeability determines how much of that gradient it can use.

06 / Vasa recta preserve the medullary gradient

Lecture PDF p. 9



Original lecture image, including visible teaching annotations (PDF p. 9).

Direction in vasa recta	Passive exchange
Descending into medulla	Blood encounters higher interstitial osmolality → water leaves blood; NaCl / other solute enters blood.
Ascending out of medulla	Blood encounters lower osmolality → solute leaves blood and water re-enters.
Hairpin arrangement	Countercurrent exchange minimises net removal / washout of medullary solute while taking reabsorbed water away.
Blood-flow figure	Medullary blood flow is LOW: less than 5% of total renal blood flow in the slide.

Three different roles

Loop of Henle: builds the gradient.
Vasa recta: preserves the gradient.
Collecting duct: uses the gradient when water-permeable.
Vasa recta do NOT create medullary hyperosmolarity.

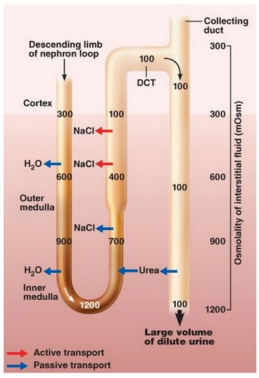
[Extension] Excessive medullary flow can wash out the gradient; very low flow also has oxygen-delivery consequences. The immediate teaching point is conservation of the solute gradient, not zero exchange of material.

07 / Collecting duct: dilute versus concentrated urine

Lecture PDF pp. 10–11

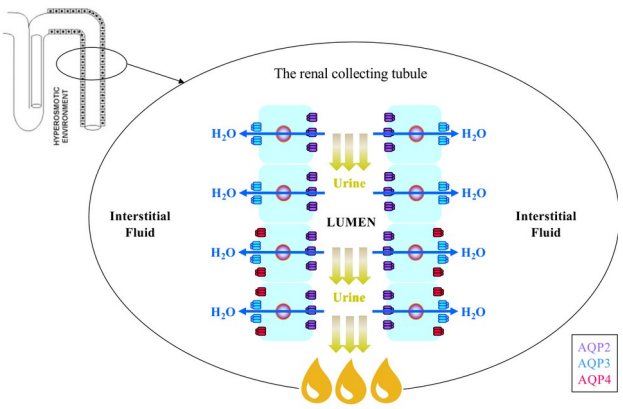
- Production of a dilute urine in the collecting duct is accomplished by simply allowing filtrate from the Loop of Henle to pass on to the renal pelvis

Overhydration
(Plasma hypo-osmolality)
↓
No vasopressin (or anti-diuretic hormone [ADH]) stimulation
↓
No water reabsorption from the lumen
↓
Large volume of diluted urine



Original lecture figure, PDF p. 10.

- Formation of a concentrated urine in the collecting ducts occurs in response to the insertion of water channel, aquaporin 2 (AQP2), onto the apical membrane of collecting duct cells, hence making them permeable to water → small amount of concentrated urine



Original lecture figure, PDF p. 11.

State	Duct permeability and urine outcome
Overhydration / plasma hypo-osmolality	VP drive decreases → little apical AQP2 → low collecting-duct water permeability → a larger volume of dilute urine.
Source's simplified chain	"No vasopressin stimulation → no water reabsorption from the lumen": applies to the regulated collecting-duct component, not all nephron water reabsorption.
Concentrated urine	AQP2 is inserted into the apical membrane → collecting-duct cells become water-permeable → water follows medullary gradient → smaller volume of concentrated urine.
Basolateral exit	AQP3 and AQP4 provide routes out of the cell toward interstitial fluid; they are labelled separately from apical AQP2.

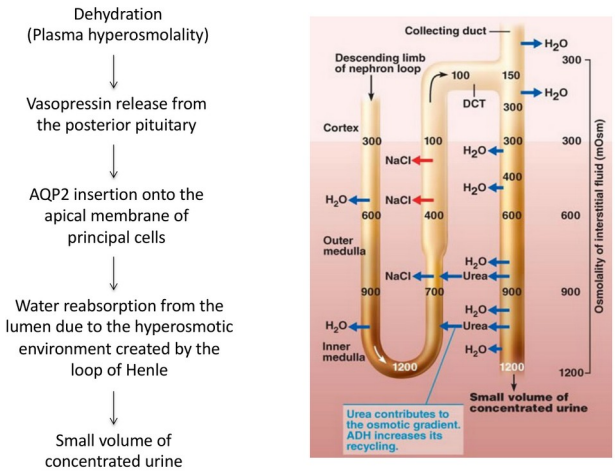
Trace water across the principal cell

Lumen → APICAL AQP2 → cell → BASOLATERAL AQP3 / AQP4
→ medullary interstitial fluid → blood.
AQP2 changes permeability; it does not actively pump water.

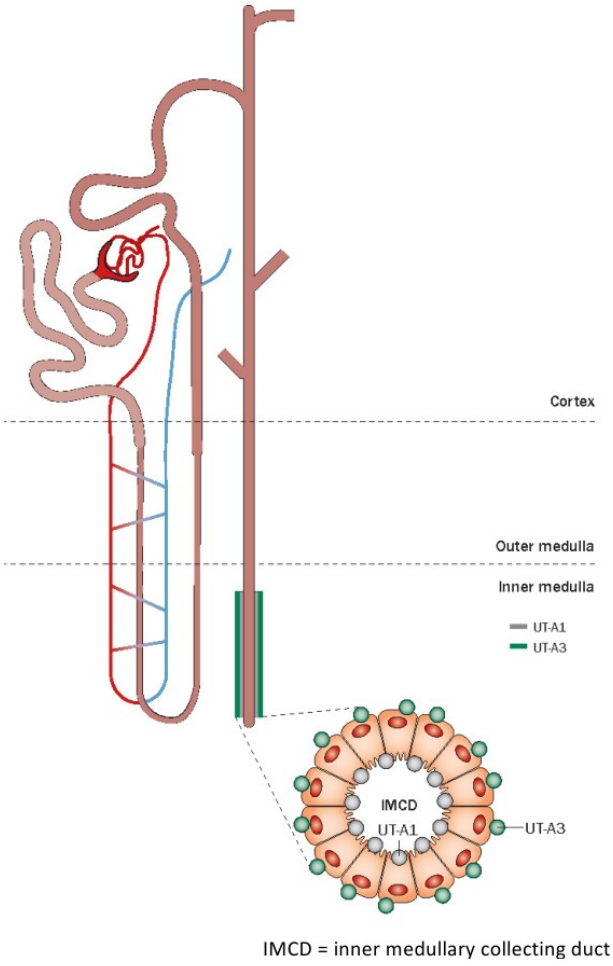
The loop has already delivered dilute fluid. Merely having a salty medulla does not force water through a water-impermeable collecting duct. Conversely, a permeable duct needs a functioning medullary gradient to concentrate effectively.

08 / Dehydration and the role of urea

Lecture PDF pp. 12–13



Original lecture figure, PDF p. 12.



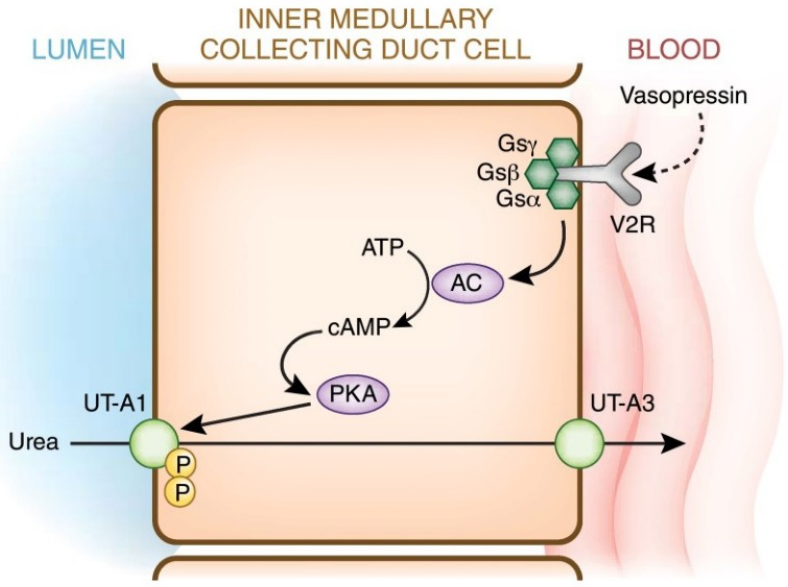
Original lecture figure, PDF p. 13.

Complete source step	Meaning
Dehydration → plasma hyperosmolality	Promotes VP release from posterior pituitary.
VP → AQP2	AQP2 appears at the apical membrane of collecting-duct PRINCIPAL cells.
Hyperosmotic surroundings	Loop-generated medullary environment draws water from lumen → small volume of concentrated urine.
Urea source / elimination	Principal nitrogenous waste of protein metabolism; eliminated almost exclusively through kidneys in urine in the lecture's description.
Concentration within duct	As water is reabsorbed, urea remaining in tubular fluid becomes more concentrated.
IMCD urea exit	In INNER MEDULLARY collecting duct, urea diffuses DOWN its concentration gradient through UT-A1 / UT-A3 → medullary interstitium.
Concentrating contribution	Medullary urea accumulation supplies part of the osmotic driving force extracting water from collecting ducts.

IMCD = inner medullary collecting duct. UT-A1 is apical; UT-A3 is basolateral in the later signalling figure. Urea is a nitrogenous waste AND a useful component of concentrating physiology; passive urea movement is distinct from ATP-driven salt transport.

09 / VP controls urea permeability: classroom Activity 3

Lecture PDF p. 14 | Original Photo 3 integrated with this lecture's urea / AQP2 mechanisms



Original lecture image, including visible teaching annotations (PDF p. 14).

Written / diagram detail	Full mechanism
Fast VP effect	Increases IMCD urea permeability by phosphorylation and apical membrane accumulation of UT-A1.
Signal in figure	Blood-side VP → V_2 receptor → Gs → adenylyl cyclase → cAMP → PKA → UT-A1 phosphorylation / trafficking.
Across-cell route	Luminal urea → UT-A1 → cell → UT-A3 → interstitial / blood side.
Overall effect	Establishes / maintains high medullary urea → supports VP-dependent urine concentration.

Original Activity 3 | Transcribed from classroom photo

What happens if we inject a small amount of hypertonic saline into the body? What is the immediate effect on urine output?

Answer approach:

- The photographed answer is a small volume of concentrated urine.
- Small hypertonic load → hyperosmolality, with relatively little direct volume loading in the question's intended comparison.
- PVN / SON activation → posterior-pituitary VP release ↑ → apical AQP2 ↑ → renal water reabsorption ↑.
- UT-A1-mediated urea handling supports the medullary gradient → concentrating ability increases.
- Urine flow falls and urine concentration rises after the hormonal / renal response develops.

Both parts of the photographed answer are retained: AQP2 enables water entry; UT-A1 supports the urea contribution to the inner-medullary gradient. Small hypertonic load is the stated condition; the effect is not a general prediction for every saline dose, time point or urine already in the bladder.

10 / Integrate the concentrating mechanism

Lecture PDF p. 15 | Summary of every written point; one self-written mechanism question

Source summary point	Retained explanation
Corticomedullary gradient	Osmolality increases from corticomedullary boundary toward the papillary tip.
Outer medulla	NaCl is the major gradient constituent; active salt reabsorption in thick ascending limbs is central.
Inner medulla	Both NaCl AND urea contribute; passive urea transport into interstitium is important.
Countercurrent arrangement	Nephron segments and vessels are arranged in opposite-flow configurations. Multiplication and exchange have different jobs.
Urine concentration	Requires interaction among medullary tubules, collecting-duct permeability, urea handling and the medullary circulation.

Integrated practice 1 | Self-written; not an original exam question

NKCC2 is blocked while VP release and AQP2 trafficking remain intact. Can high VP alone fully restore urine concentration? Predict the effect of increased vasa-recta washout as well.

Answer approach:

• NKCC2 blockade reduces ascending-limb salt removal → medullary gradient weakens; tubular fluid is also less effectively diluted.

• AQP2 still raises collecting-duct water permeability, but a weaker osmotic driving force limits water extraction.

• High VP alone cannot substitute for the lost gradient; urea may still contribute without fully replacing outer-medullary salt handling.

• Increased medullary washout further dissipates the gradient, even though the vasa recta were never its primary generator.

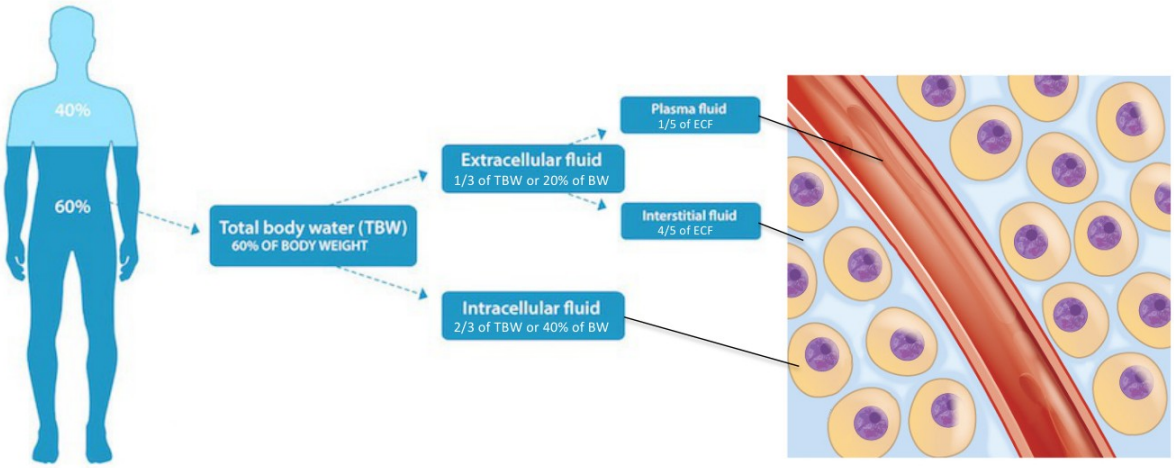
[Extension] This applies the bumetanide-sensitive transporter already named on slide 5. It is a self-written integrated exercise, not a new original class question.

11 / Body-fluid compartments and source percentages

Lecture PDF p. 17

- A reduction in body fluids can have major effects on the body:
  - ➢ 3% reduction → thirst
  - ➢ 8% reduction → illness
  - ➢ 10% reduction → death

It is very essential to maintain a homeostasis of the volume and composition of bodily fluids



Original lecture image, including visible teaching annotations (PDF p. 17).

Compartment	Fraction of total body water (TBW)	Fraction of body weight (BW)
TBW	100%.	About 60% in the adult teaching model.
Intracellular fluid (ICF)	2/3 of TBW.	40% of BW.
Extracellular fluid (ECF)	1/3 of TBW.	20% of BW.
Plasma	1/5 of ECF.	About 4% of BW, derived from the slide fractions.
Interstitial fluid	4/5 of ECF.	About 16% of BW, derived from the slide fractions.

Written dehydration figures — retain with their limitation

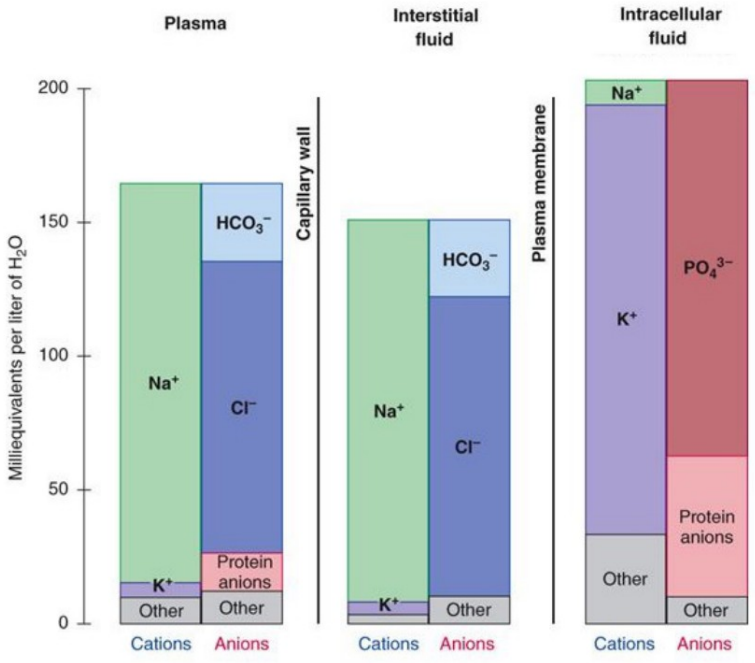
The source lists reduction in body fluids: 3% → thirst; 8% → illness; 10% → death.

These illustrate escalating severity, not guaranteed clinical thresholds. The slide does not clearly define the denominator for these losses; do not silently relabel them as percentage loss of body weight.

Maintain BOTH fluid volume and composition to protect cells. Approximate compartment fractions vary with age, sex and adiposity; do not combine the later cartoon's "70% water" with this 60%-TBW calculation as if they were the same quantitative model.

12 / Major ions and what high osmolality means

Lecture PDF pp. 18, 23



Original lecture image, including visible teaching annotations (PDF p. 18).

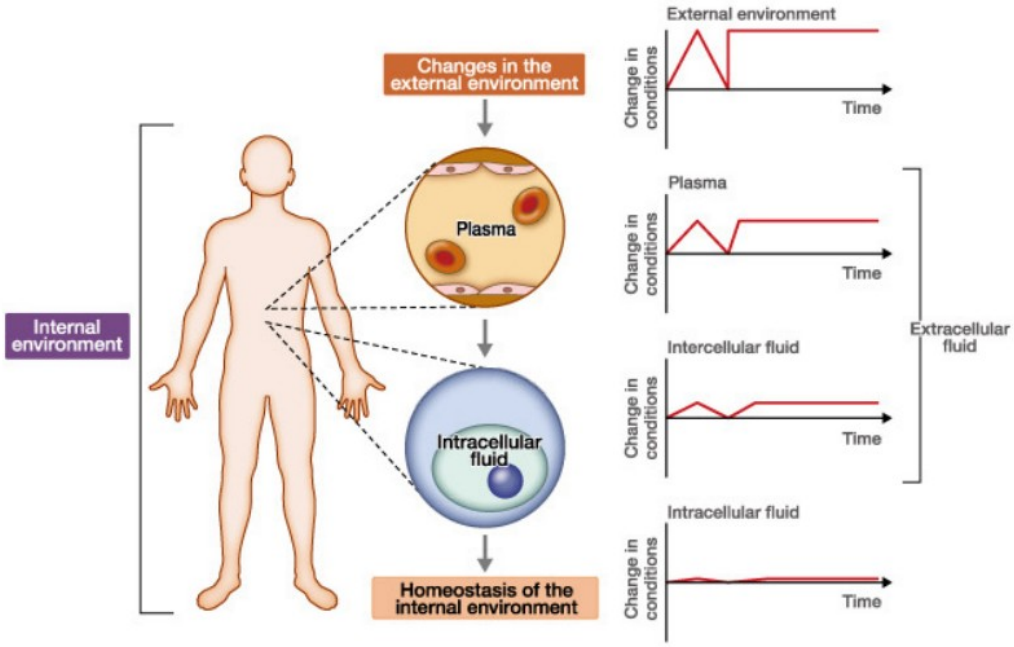
Compartment	Predominant positive ions	Predominant negative components
Plasma	Na^+; much smaller K^+ contribution.	Cl^-, HCO_3^- and negatively charged plasma proteins.
Interstitial fluid	Na^+ dominates.	Cl^- and HCO_3^-; much less protein than plasma.
Intracellular fluid	K^+ dominates; Mg^{2+} contributes.	Phosphate and protein anions dominate.

Term	Definition / significance
Osmolarity	Osmotically active particles per litre of solution; mOsm/L.
Osmolality	Particles per kilogram of WATER; mOsm/kg H_2O. Numerically close to osmolarity in dilute body fluids.
High value	More solute particles relative to water; it describes concentration, not fluid volume or total-body solute alone.
Osmotic water movement	Across a water-permeable membrane, a difference in effectively non-permeant solutes drives water toward the more concentrated side.
Tonicity [Extension]	Effect on cell volume depends on effective, non-penetrating osmoles; total osmolality alone does not describe every solute's cell-volume effect.

The source calls osmolality "total amount of solutes" and says it is controlled "purely" by free water. Correction: it is a CONCENTRATION and can change through solute or water changes. The VP / thirst axis chiefly adjusts water; that does not make solute changes impossible. The ionic chart uses mEq/L water, an electrical-equivalent scale, not the same unit as mOsm/kg.

13 / Buffering fluid change and coordinating responses

Lecture PDF p. 19



Original lecture image, including visible teaching annotations (PDF p. 19).

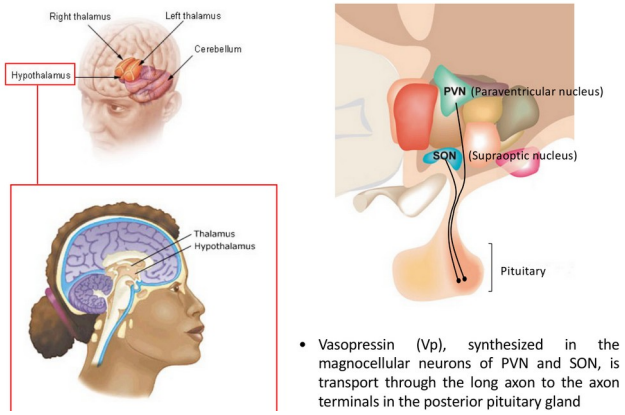
Source statement	Retained meaning
Sequence of buffering	A disturbance is first buffered by plasma, then dampened by interstitial fluid and finally by intracellular fluid in the source model.
Ultimate target	Maintenance of a stable INTRACELLULAR environment.
Physiological control centres	Brainstem, lamina terminalis and hypothalamo-pituitary axis.
Physiological outputs	VP / ADH; renin–angiotensin–aldosterone system.
Behavioural responses	Circumventricular organs and renin–angiotensin signalling participate in thirst / fluid intake.

Do not reduce homeostasis to one variable

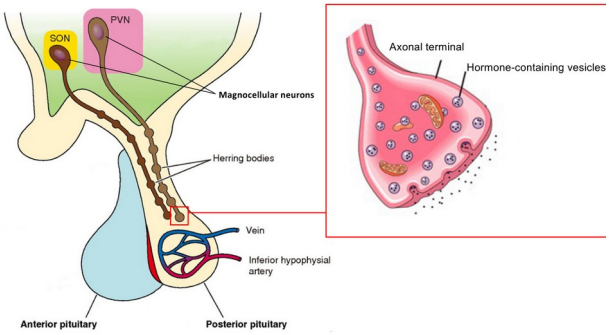
Fluid volume, sodium concentration, total solute content and blood pressure are related but distinct.
A disturbance can activate renal / vascular physiology AND drinking behaviour.
The source graph shows progressively dampened fluctuations, rather than claiming that ICF never changes.

14 / VP synthesis, transport and storage

Lecture PDF pp. 20–21



Original lecture figure, PDF p. 20.



Original lecture figure, PDF p. 21.

Stage	All written details
Synthesis	VP is synthesised in MAGNOCELLULAR neurons in hypothalamic PVN and SON.
PVN	Paraventricular nucleus.
SON	Supraoptic nucleus.
Axonal transport	VP travels down long axons to terminals in the POSTERIOR pituitary.
Storage	Hormone-containing secretory vesicles occur in axons / axonal terminals until mobilised for secretion.
Herring bodies	Source calls the secretory granule a Herring body. More precisely, Herring bodies are neurosecretory axonal swellings containing accumulated granules.

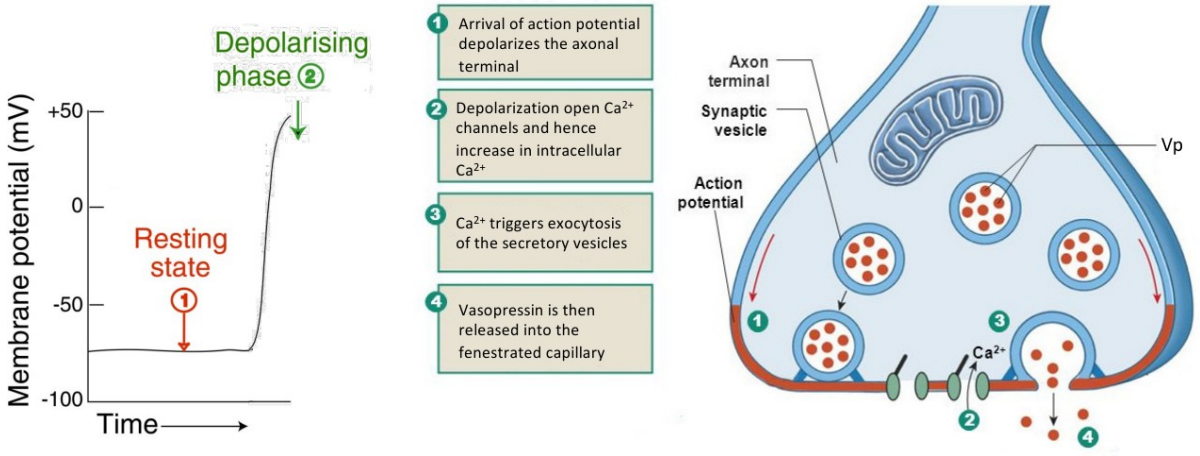
Separate production from release

Hypothalamic cell body → synthesis.
Axon → transport and storage.
Posterior-pituitary terminal → release into blood.
The posterior pituitary does not synthesise VP as a separate endocrine cell population.

MNC = magnocellular neurosecretory neuron. The storage figure also labels anterior / posterior pituitary, vein and inferior hypophysial artery; the posterior-pituitary circulation receives the released hormone.

15 / A stimulus becomes circulating vasopressin

Lecture PDF p. 22



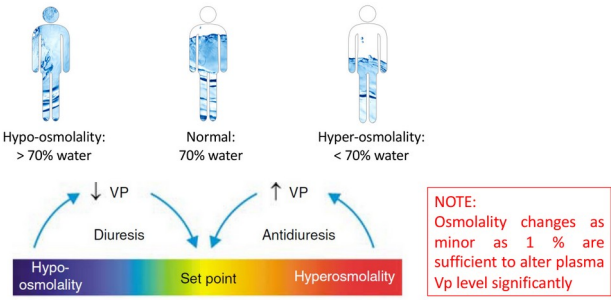
Original lecture image, including visible teaching annotations (PDF p. 22).

Step	Complete source mechanism
1. Detect / integrate stimulus	Signals are detected or integrated at the magnocellular cell bodies in hypothalamic SON / PVN.
2. Electrical response	Appropriate stimulus depolarises neurons → action potentials propagate down their axons.
3. Terminal depolarisation	Arriving action potential depolarises the axon terminal.
4. Calcium entry	Voltage-dependent Ca^{2+} channels open → intracellular Ca^{2+} increases.
5. Exocytosis	Ca^{2+} triggers fusion / exocytosis of VP-containing secretory vesicles.
6. Blood entry	VP enters the fenestrated capillaries at the posterior pituitary.

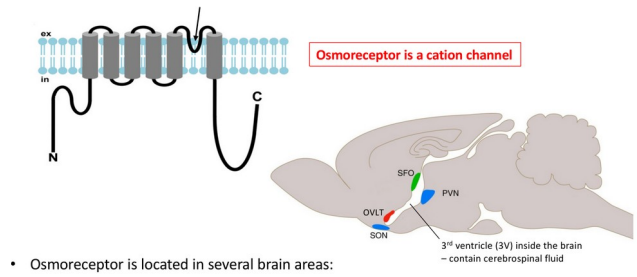
The source figure includes resting and depolarising phases of membrane potential. Releasing VP requires calcium-dependent vesicular exocytosis; transporting hormone down the axon and releasing it into circulation are separate steps.

16 / Osmotic sensing: sensitivity and brain sites

Lecture PDF pp. 23–24



Original lecture figure, PDF p. 23.



Original lecture figure, PDF p. 24.

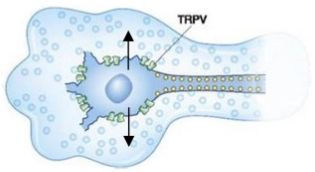
Written / figure point	Detail retained and clarified
Sensitivity	A change in osmolality as small as 1% can significantly change plasma VP.
Negative feedback	Hyperosmolality → VP ↑ → antidiuresis; hypo-osmolality → VP ↓ → diuresis.
Water cartoons	Source illustrates normal 70%, >70% for hypo-osmolality and <70% for hyperosmolality. These are schematic relative-water cartoons, not a universal quantitative definition of normal TBW.
Receptor model	Mechanosensitive cation-channel / TRPV-family osmosensory mechanism converts osmotic cell-volume changes to electrical signals. Pore-loop segment is the cation-conducting pore.
Locations named	OVLT; SFO; hypothalamic PVN and SON.
OVLT / SFO	Organum vasculosum of the lamina terminalis / subfornical organ; circumventricular organs (CVOs) exposed to circulating ionic / hormonal signals through specialised barrier anatomy.
Third ventricle	3V contains cerebrospinal fluid.

Qualification: OVLT and SFO are sensory CVOs with a weak / absent conventional blood–brain barrier. Do not extend that label to all PVN / SON tissue. TRPV mechanisms are part of the teaching model; an osmoreceptor is an osmosensitive neuron / sensing apparatus, not necessarily one universal receptor protein in every site.

17 / Hypertonicity versus hypotonicity: electrical chain

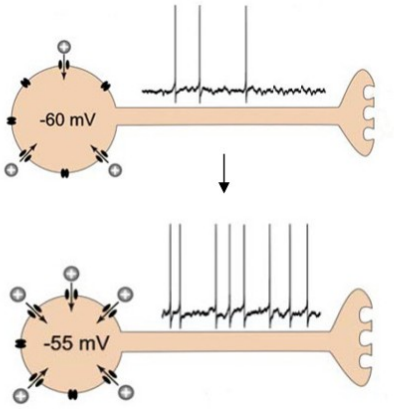
Lecture PDF p. 25

1. Dehydration/ plasma hyperosmotically induce cell shrinkage

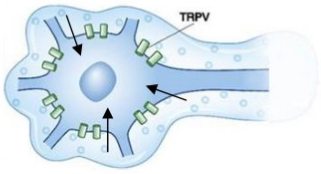


Water outflux from the cell by osmosis

2. Cell shrinkage causes a conformational change of osmoreceptor (mechano-sensitive cation channel), leading to opening of the pore loop and hence +ve ions influx → depolarization and firing of action potential along the axon to the axonal terminal

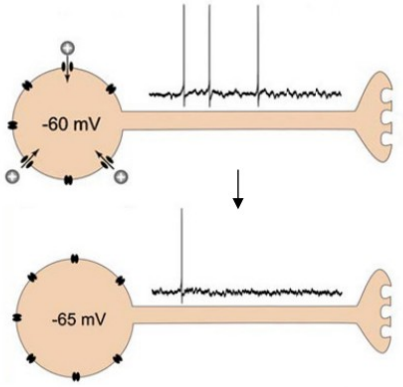


1. Systemic hypotonicity causes entry of water into the cells



Water influx into the cell by osmosis

2. Stretching of cell membrane suppresses osmoreceptor, reducing whole-cell cation conductance → membrane hyper-polarization and inhibits firing of action potential



Original lecture image, including visible teaching annotations (PDF p. 25).

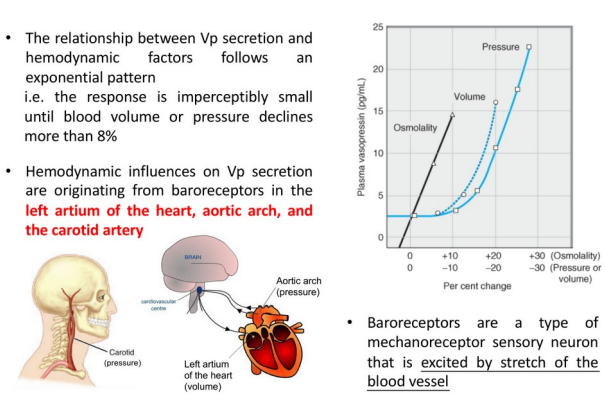
Step	Hyperosmolality / dehydration	Hypo-osmolality / excess water
Osmotic water shift	Water leaves osmosensitive cells.	Water enters cells.
Cell volume	Shrinkage.	Swelling / membrane stretch.
Mechanosensitive conductance	Source model: channel conformational change / pore opening → cation influx ↑.	Source model: cation conductance suppressed.
Electrical state	Depolarisation; illustrated change from about −60 toward −55 mV.	Hyperpolarisation / reduced depolarisation.
Firing / VP	Action-potential firing ↑ → VP release ↑.	Firing inhibited → VP release ↓.
Renal consequence	Water conservation; helps restore osmolality.	Water excretion; helps restore osmolality.

Water movement and channel response are sequential

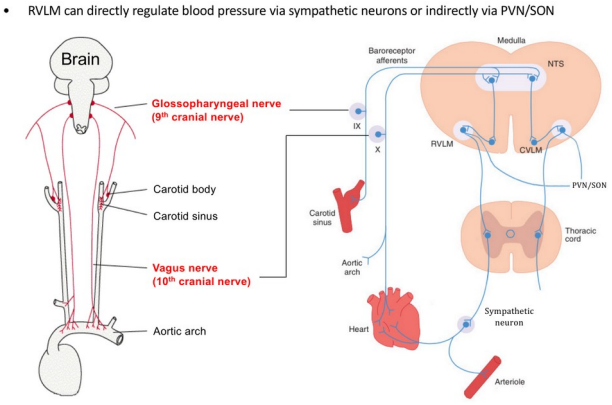
Osmotic water shift → cell-volume / membrane-mechanical change
→ altered cation conductance → changed membrane potential
→ changed firing → changed hormone release.
Do not skip directly from “salty plasma” to “water channel opens in the brain”.

18 / Volume and pressure signals: different sensitivity

Lecture PDF pp. 26–27



Original lecture figure, PDF p. 26.



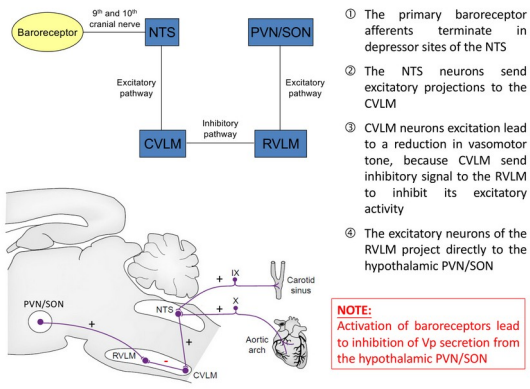
Original lecture figure, PDF p. 27.

Written point	All named sites / relationships
Response shape	Haemodynamic influence on VP is highly nonlinear / approximately exponential in the source graph.
Magnitude threshold	Source says response is small until blood volume or pressure declines by more than about 8%.
Volume-sensitive site	Left atrial stretch receptors.
Pressure-sensitive sites	Aortic arch and carotid-sinus arterial baroreceptors.
Receptor type	Mechanoreceptor sensory neurons excited by vessel / wall stretch.
Afferent route	Carotid-sinus signals via glossopharyngeal CN IX; aortic signals via vagus CN X → NTS in medulla.
Central relay	NTS → CVLM → RVLM; RVLM can regulate pressure via sympathetic neurons and influence hypothalamic PVN / SON.

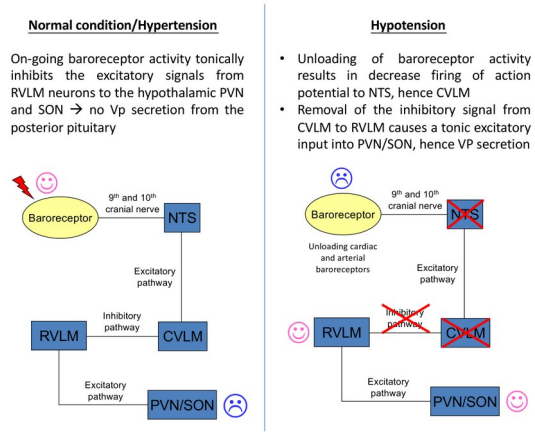
Compare source sensitivities: ~1% osmotic change versus a larger ~8% haemodynamic decline. They are approximate teaching thresholds, not hard cutoffs in every person. Carotid BODY is principally a chemoreceptor organ; carotid SINUS contains the pressure receptors relevant here. NTS = nucleus tractus solitarius / nucleus of the solitary tract; CVLM / RVLM = caudal / rostral ventrolateral medulla.

19 / Baroreflex circuit and classroom cardiovascular effects

Lecture PDF pp. 28–29 | Photos 1–2 supplement this exact lecture pathway



Original lecture figure, PDF p. 28.



Original lecture figure, PDF p. 29.

Connection in the slide	Sign and net consequence
Baroreceptor → NTS	Primary afferents terminate at depressor regions.
NTS → CVLM	Excitatory.
CVLM → RVLM	Inhibitory; CVLM excitation lowers vasomotor drive.
RVLM → PVN / SON	Excitatory; reduced RVLM drive tends to reduce VP output.
Normal / hypertension	Ongoing baroreceptor firing provides a tonic inhibitory influence on the VP pathway.
Hypotension / unloading	Afferent firing ↓ → NTS / CVLM activation ↓ → RVLM disinhibited → hypothalamic excitatory drive ↑ → VP ↑.

Original supplemental prompt | Transcribed from classroom photo

With ongoing baroreceptor activity under normal conditions / hypertension, what would happen to the cardiovascular system?

Answer approach:

- Sympathetic drive to blood vessels ↓ → vasodilation → vascular resistance and BP tend to fall.
- Sympathetic drive to the heart ↓ → ventricular inotropy / contractility ↓.
- Parasympathetic cardiac activity ↑ → heart rate ↓ (including reduced pacemaker funny-current drive) and conduction velocity ↓.
- Cardiac output ↓; vasodilation also lowers BP.
- VP inhibition → less renal water retention → larger volume of more dilute urine → blood volume tends to fall.
- Net teaching outcome: restoration of body-fluid and pressure homeostasis.

The PPT's "no VP secretion" depicts the inhibitory branch, not universal absence of basal VP at normal BP. Osmotic and haemodynamic inputs are integrated. The photo's "NTS is the CVC" is shorthand: NTS is part of a broader medullary cardiovascular-control network. The cartoon wiring is the lecture's simplified model rather than a complete map of all hypothalamic connections.

20 / Bilateral carotid occlusion and vagotomy

Lecture PDF pp. 26–29 | Classroom Photo 4: original Activity 4 and follow-up

Original Activity 4 | Transcribed from classroom photo

If implanted pneumatic cuffs perform bilateral carotid occlusion in an experimental animal, what changes would occur in urine output?

Answer approach:

The photo's proposed chain is: carotid blockage → aortic BP ↑ → aortic-arch baroreceptors activated → CN X firing to NTS / CVLM ↑ → RVLM and PVN / SON suppressed → VP ↓ → urine output ↑.

This is an AORTIC-DOMINANT interpretation that assumes the rise in aortic pressure provides the dominant inhibitory input.

Competing input	Conditional physiological prediction
Carotid-sinus unloading	Occlusion BELOW the sinus reduces sinus pressure → CN IX firing ↓ → less reflex inhibition → sympathetic / VP drive tends to rise.
Higher aortic pressure	If systemic / aortic pressure rises sufficiently, aortic-arch firing via CN X rises → opposes carotid unloading.
Photo's proposed answer	Aortic inhibition dominates → VP ↓ → urine flow ↑. Retain as the answer under that assumed state.
General answer	Net result depends on cuff location, sinus / aortic pressures and intact nerves; occlusion alone does not guarantee diuresis.

Original follow-up | Transcribed from classroom photo

What happens if we perform vagotomy in this case?

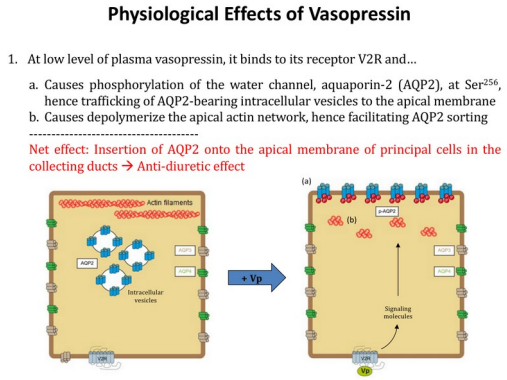
Answer approach:

- In the photo's setup, cutting vagal afferents removes the aortic-arch inhibitory signal.
- Relative to the intact-vagus, aortic-dominant state: VP drive can rise → water reabsorption ↑ → urine flow ↓.
- CN IX carotid-sinus afferents are not cut by vagotomy. With carotid unloading still present, removing aortic feedback can strengthen the low-pressure / sympathetic response.
- Vagotomy also affects other vagal functions; this question isolates the baroreceptor–VP component.

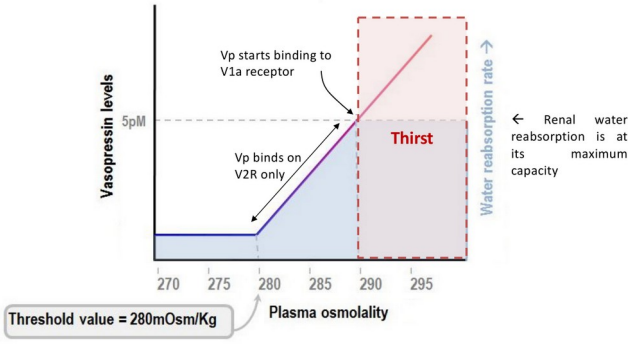
State both sensors, both nerves and the assumed pressure changes. In the photographed aortic-dominant state, vagotomy removes aortic inhibitory feedback and can increase VP relative to the intact-vagus state. It does not cut glossopharyngeal carotid-sinus afferents. Renal urine formation changes after the physiological response develops; urine already in the bladder does not instantly change composition.

21 / Low VP: V_2, AQP2 trafficking and thirst threshold

Lecture PDF pp. 31–32



Original lecture figure, PDF p. 31.



Original lecture figure, PDF p. 32.

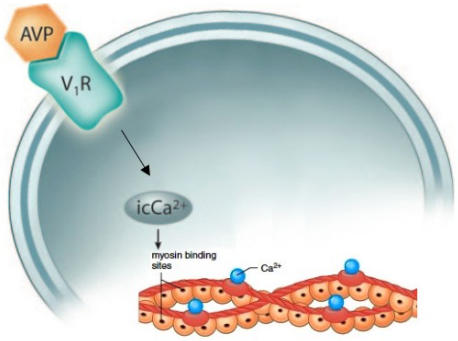
Written detail	Mechanism
Low plasma VP	Renal V_2 receptors mediate antidiuresis.
AQP2 phosphorylation	AQP2 is phosphorylated at Ser^{256} → AQP2-bearing vesicles traffic to apical membrane.
Actin remodelling	Depolymerisation of apical actin network facilitates AQP2 sorting / delivery.
Net cellular effect	Apical AQP2 in collecting-duct principal cells ↑ → water permeability ↑ → antidiuresis.
Threshold graph	Source shows VP response starting around 280 mOsm/kg.
Thirst	Source says thirst is initiated above about 290 mOsm/kg, once maximal renal VP water-retaining effect is reached at about 5 pM plasma VP.
Further VP rise	Additional pressor / V_1a effects become increasingly important; the renal water-reabsorption curve has approached its plateau.

[Extension] Renal V_2 signalling uses Gs–cAMP–PKA. The graph's "V_2 only / V_1a starts" separation is schematic, not absolute receptor exclusivity or a universal exact concentration threshold. Thirst is subjective drive to drink, adding behavioural intake when renal water conservation alone is insufficient.

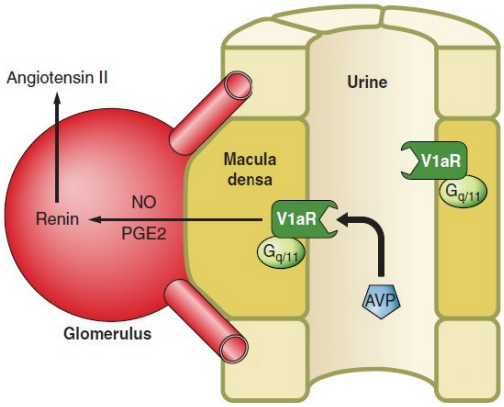
22 / Higher VP: V_1a vessels and the renin link

Lecture PDF p. 33

- V1aR is expressed on vascular smooth muscle cells of the systemic and coronary circulations
- Activation of V1aR leads to increase in intracellular Ca^2+, hence activating myosin light chain kinase → smooth muscle contraction and therefore vasoconstriction



- V1aR is expressed on Macula Densa cells
- Activation of V1aR leads to the production of nitric oxide (NO) and prostaglandin E2 (PGE2) by macula densa cells, which subsequently stimulate renin production by the juxtaglomerular cells located on the afferent arterioles



Original lecture image, including visible teaching annotations (PDF p. 33).

Route	All source steps
Direct vascular route	V_1a receptors on systemic AND coronary vascular smooth muscle → intracellular Ca^{2+} ↑ → myosin light-chain kinase (MLCK) activated → muscle contracts → vasoconstriction.
Indirect route taught in lecture	V_1a receptors on macula-densa cells → nitric oxide (NO) and prostaglandin E_2 (PGE_2) production → renin production / release from JG granular cells on afferent arterioles.
Later amplification	Renin–angiotensin system links to aldosterone, thirst, further VP secretion and vasoconstriction.

Keep the cell types and mediators separate

Vessel: V_1a → Ca^{2+} → MLCK → contraction.

Macula densa, in the source route: V_1a → NO / PGE_2 → neighbouring JG-cell renin.

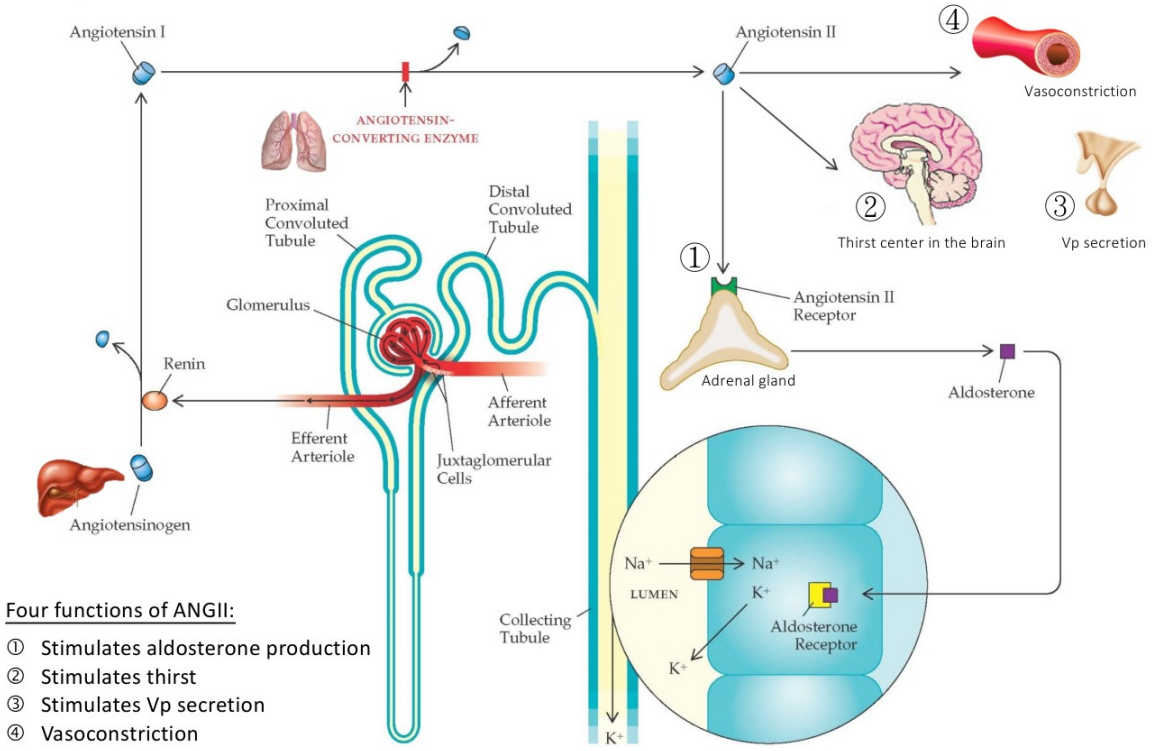
JG cell is not the same cell as macula densa.

Qualification: retain this VP–macula-densa route as the lecture's mechanistic model, not a universal prediction that VP always raises net circulating renin. Renin integrates perfusion, NaCl delivery and sympathetic inputs; VP's pressure / volume effects can create opposing feedback. MLCK phosphorylates myosin light chains during smooth-muscle contraction.

23 / Renin–angiotensin cascade and all four outputs

Lecture PDF p. 34 | Also links the integrated trigger diagram on p. 42

- Renin catalyzes the conversion of a plasma protein called angiotensinogen into angiotensin I, which is then further convert by angiotensin-converting enzyme (ACE) into angiotensin II (ANGII)



Original lecture image, including visible teaching annotations (PDF p. 34).

The enzymatic cascade

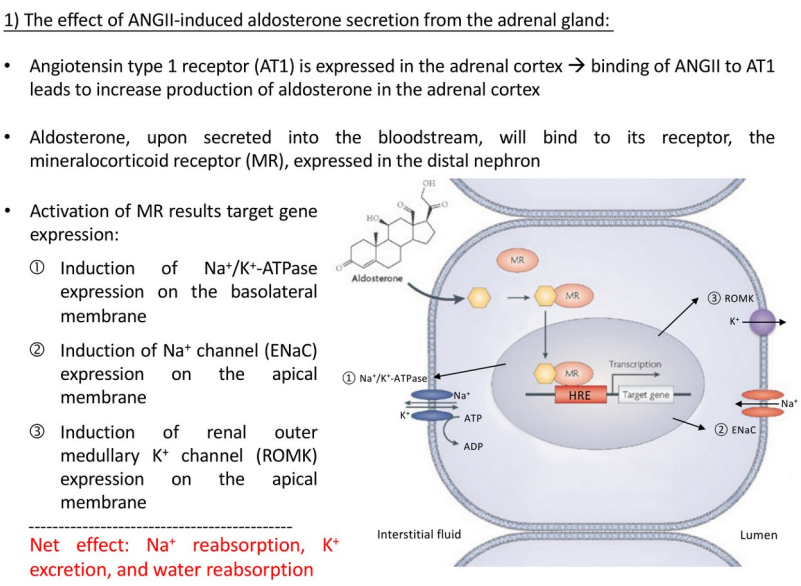
Liver-derived angiotensinogen —renin→ angiotensin I
—angiotensin-converting enzyme (ACE)→ angiotensin II (ANGII).
Renin comes from renal juxtaglomerular granular cells.

ALL four ANGII functions	Primary target / later detail
1. Aldosterone production	Adrenal cortex → distal-nephron sodium / potassium handling; notes p. 24.
2. Thirst	SFO / OVLT → MnPO → cortical circuits → drinking; p. 25.
3. VP secretion	SFO / OVLT → MnPO → magnocellular PVN / SON → posterior pituitary; p. 25.
4. Vasoconstriction	Vascular AT_1 plus central sympathetic routes; p. 26.

The final flowchart adds reduced afferent-arteriolar stretch, reduced NaCl delivery sensed at the macula densa and sympathetic activation as renin-stimulatory signals. Its wording “NaCl concentration in ascending limb” should be read at the macula-densa / end-thick-ascending-limb sensing region. ACE is prominent on vascular endothelium including pulmonary vessels; it is not confined to the lungs.

24 / ANGII → aldosterone → three transport proteins

Lecture PDF p. 35



Original lecture image, including visible teaching annotations (PDF p. 35).

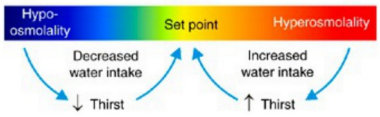
Stage	All written details
ANGII receptor	AT_1 in ADRENAL CORTEX → aldosterone production increases.
Hormone transport / target	Aldosterone enters blood → binds mineralocorticoid receptor (MR) in distal nephron.
Gene regulation	Activated MR alters target-gene expression; figure labels HRE, hormone-response element.
1. Basolateral Na^+/K^+-ATPase	Expression increases → Na^+ moves into interstitium and K^+ into cell; uses ATP.
2. Apical ENaC	Epithelial Na^+ channel expression increases → luminal Na^+ enters cell.
3. Apical ROMK	Renal outer-medullary K^+ channel expression increases → K^+ secreted into lumen.
Net effect	Na^+ reabsorption ↑; K^+ excretion ↑; associated water retention helps restore volume.

Lumen and interstitial sides are explicitly different. Aldosterone does not insert AQP2; water retention depends on available water and the segment's water permeability / VP state. [Extension] MR is an intracellular steroid receptor; its transcriptional actions differ from rapid VP-controlled vesicle trafficking.

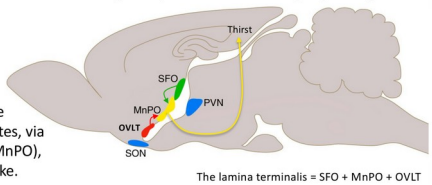
25 / ANGII: drinking and further VP release

Lecture PDF pp. 36–37

- Thirst is a subjective perception that provides the urge to drink fluids. It is a component of the regulatory mechanisms that maintain body fluid homeostasis and ultimately is essential for survival.

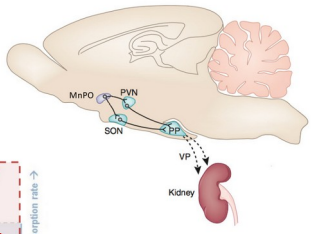


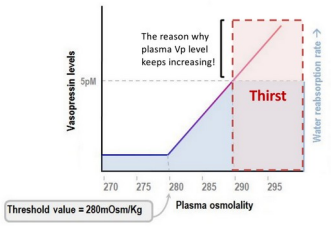
- SFO and OVLT are the CVOs that detect changes in plasma ANGII level by expressing AT1 receptor.
- By responding to ANGII and integrating plasma osmolality information, these regions then relay the signals to various cortical sites, via median preoptic nucleus (MnPO), to drive thirst and fluid intake.



Original lecture figure, PDF p. 36.

- SFO and OVLT are the CVOs that detect changes in plasma ANGII level by expressing AT1 receptor
- Once the SFO and OVLT neurons are electrophysiologically activated, signals will be send via MnPO to the magnocellular neurons of the hypothalamic PVN & SON





Original lecture figure, PDF p. 37.

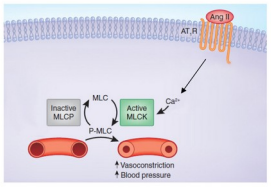
Written point	Retained pathway
Thirst definition	Subjective perception generating the urge to drink; part of homeostasis and essential for survival.
ANGII detection	SFO and OVLT are sensory CVOs expressing AT_1 receptors; they detect circulating ANGII.
Integration	ANGII signals are integrated with plasma-osmolality information.
Drinking branch	Signals relay through MEDIAN PREOPTIC NUCLEUS (MnPO) to cortical sites → thirst and fluid intake.
Lamina terminalis	Source circuit: SFO + MnPO + OVLT.
VP branch	Electrophysiologically activated SFO / OVLT → MnPO → MAGNOCELLULAR neurons in PVN / SON → VP secretion.
Why VP can keep rising	Further endocrine / neural drive continues after the kidney's water-retaining response is close to maximal; vascular and intake responses are additional outputs.

Retain the source's nucleus names: median PREOPTIC nucleus, not a posterior-pituitary nucleus. Magnocellular PVN / SON cells project to the posterior pituitary; the parvocellular PVN branch to RVLM is a separate route on p. 26.

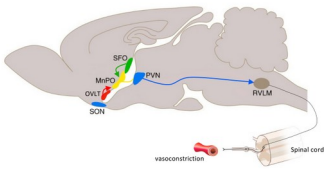
26 / ANGII vasoconstriction: direct and central

Lecture PDF pp. 38–39

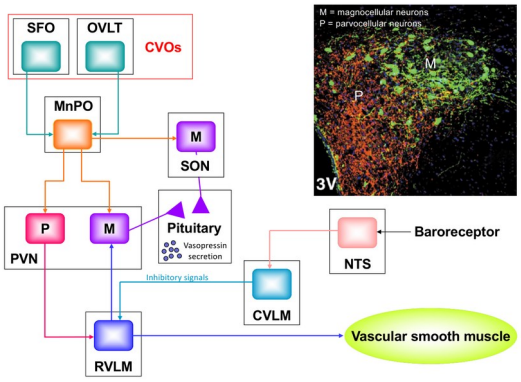
- Binding of ANGII to AT1 receptor increases intracellular Ca^{2+}, which, activates myosin light chain kinase (MLCK) → blood vessel smooth muscle contraction and elevation of blood pressure



- Neurons within MnPO are connected to RVLM, the "pressor area" in the brainstem, via the **parvocellular neurons** of the hypothalamic PVN
- Once SFO and OVLT are activated by circulating ANGII, RVLM will send sympathoexcitatory signals to cause vasoconstriction



Original lecture figure, PDF p. 38.



Original lecture figure, PDF p. 39.

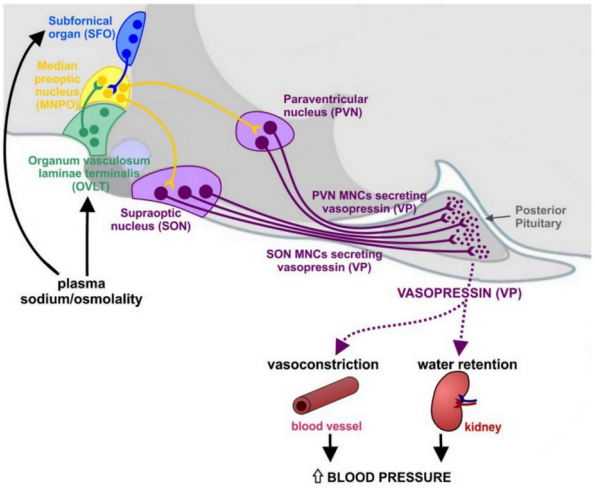
Route	All source connections
Direct vascular	ANGII → AT_1 on vascular smooth muscle → intracellular Ca^{2+} ↑ → MLCK activation → contraction → BP ↑.
Central pressor route	Circulating ANGII → SFO / OVLT → MnPO → PARVOCELLULAR PVN → RVLM → sympathoexcitatory spinal output → vascular constriction.
Endocrine route in integrated map	MnPO → magnocellular PVN / SON → posterior pituitary → VP release.
Baroreceptor brake	Baroreceptors → NTS → CVLM → inhibitory signal to RVLM; integrates pressure feedback with the ANGII circuit.
M versus P	M = magnocellular; P = parvocellular. Both are labelled in the source schematic and micrograph.

PVN is not a single-output box

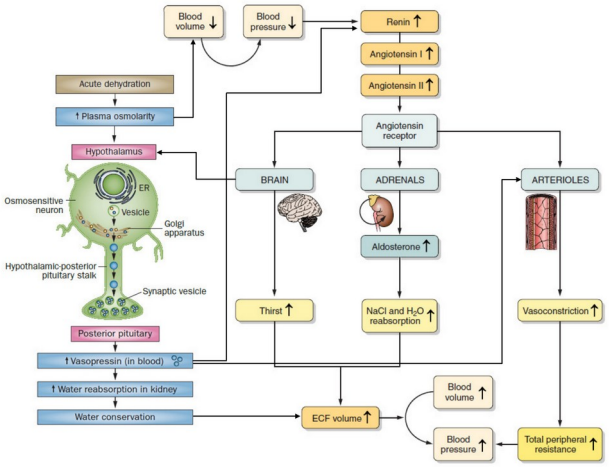
Magnocellular PVN → hormonal VP output.
Parvocellular PVN → autonomic / RVLM pressor pathway.
SON contributes magnocellular VP neurons.
Vascular AT_1 action is also direct; it does not require every ANGII effect to pass through the brain.

27 / Whole-system integration: water and pressure

Lecture PDF pp. 40–41 | Image-only slides transcribed here as mechanisms



Original lecture figure, PDF p. 40.



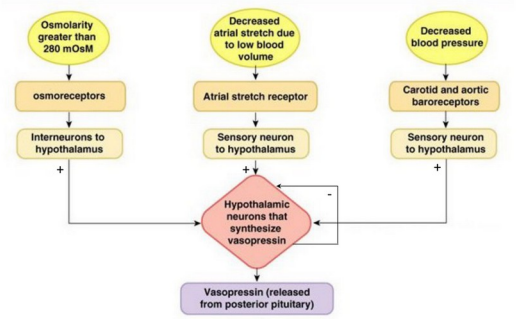
Original lecture figure, PDF p. 41.

Diagram component	Full pathway to read
Plasma sodium / osmolality	Sensory SFO / OVLT → MnPO → VP-secreting PVN / SON magnocellular neurons → posterior pituitary.
VP outputs	Kidney water retention + vascular vasoconstriction → support BP.
Acute dehydration	Plasma osmolarity ↑ and blood volume / pressure ↓ may occur together → activate hypothalamic and renin pathways.
ANGII branches	Brain thirst; adrenal aldosterone; arteriolar constriction; hypothalamic VP-related signalling.
Renal and intake outcomes	Water conservation + NaCl / water reabsorption + drinking → ECF / blood volume ↑.
Vascular outcome	Constriction → total peripheral resistance ↑; volume and resistance together support pressure restoration.

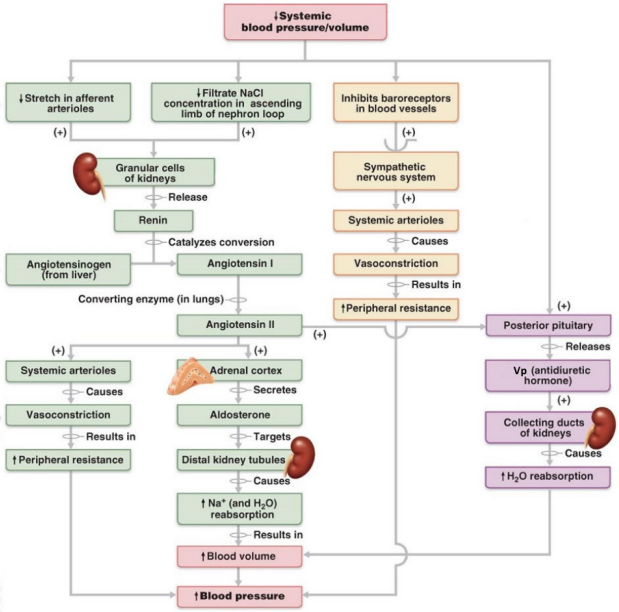
Source p. 40 also labels median preoptic nucleus, PVN / SON MNCs and posterior pituitary. Source p. 41 includes intracellular vesicles / Golgi / ER / axonal storage in its neurosecretory cartoon. The schematics combine branches; their arrows do not imply every dehydrating process changes sodium, volume and pressure by the same proportion.

28 / Integrated low-pressure flowchart and VP stimuli

Lecture PDF pp. 30, 42



Original lecture figure, PDF p. 30.



Original lecture figure, PDF p. 42.

Branch in the image-only summary	Retained causal chain
Renal pressure branch	Systemic pressure / volume ↓ → afferent-arteriolar stretch ↓ → granular-cell renin ↑.
Tubular branch	Reduced NaCl delivery sensed near macula densa → renin stimulation.
Neural branch	Arterial baroreceptor unloading → sympathetic activation → systemic arteriolar constriction and renal renin stimulation.
Hormonal cascade	Angiotensinogen from liver → renin → ANG I → ACE → ANG II.
Aldosterone branch	ANG II → adrenal cortex → aldosterone → distal tubular Na^+ retention / associated water retention → blood volume ↑.
VP branch	Posterior pituitary → VP → collecting ducts → H_2O reabsorption ↑.
Final result	Blood volume and peripheral resistance increase → arterial pressure restored toward its target.

Source p. 30 additionally lists osmolarity >280 mOsm, decreased atrial stretch due to low blood volume, and decreased arterial pressure as convergent inputs to hypothalamic VP neurons. In the osmoregulatory graph, 280 refers to the VP threshold; ~290 refers to thirst. Low NaCl is sensed by macula-densa signalling, not by a direct “salt receptor” on the renin granule itself.

29 / Diabetes insipidus and classroom Activity 6

Lecture PDF p. 43 | Photo 5: water restriction, sodium 160, urine osmolarity 500

Disease / genetics	All source details
DI definition	Kidney cannot concentrate urine adequately under appropriate physiological stimulation. Not all DI means absent hormone.
Central DI	Defect of VP production or release.
Nephrogenic DI	Kidney fails to respond adequately to VP; source examples: V_2-receptor mutation or AQP2 defect with impaired plasma-membrane trafficking.
V_2 receptor gene	X-linked; source locus Xq28; protein vasopressin type-2 receptor (AVPR2 / V_2R).
AQP2 gene	Autosomal; source locus 12q13; protein AQP2.

Original Activity 6 — case and all options | Transcribed from classroom photo

Patient: a 48-year-old woman.

• Severe polyuria (high urine output) and polydipsia (excessive drinking); no urinary glucose.

• After overnight water restriction: lethargy and confusion.

• Plasma sodium: 160 mEq/L; source normal range 135–145.

• Urine osmolarity: 500 mOsm/L; source comparison >850.

Question: most likely diagnosis?

A. Hyperaldosteronism B. Renin-secreting tumour

C. Diabetes mellitus D. Diabetes insipidus E. SIADH

Answer approach:

• Best answer among the options: D, diabetes insipidus (DI).

• Plasma Na^+ 160 indicates a marked water deficit relative to sodium. Continued polyuria and inadequate concentration during water restriction indicate impaired water conservation.

• No urinary glucose makes glucose-driven osmotic diuresis less likely in the given comparison.

• A urine value of 500 is not maximally dilute; it is INAPPROPRIATELY LOW for the hypernatraemic, water-deprived state. Partial DI / partial concentrating defects are possible.

• The data do not establish central versus nephrogenic DI.

Why the lecture says DI often has no hypernatraemia

Intact thirst + access to water can replace urinary water losses, keeping plasma sodium near normal. Water restriction, inability to drink or impaired thirst removes that compensation → hypernatraemia can develop, as in this photographed case.

The source's >850 mOsm/L is a post-restriction comparison, not a universal random-urine normal. Urine 500 is insufficiently concentrated for the sodium-160 water-deprived state, but is not maximally dilute; partial DI / other partial concentrating defects remain possible. Central versus nephrogenic disease needs more evidence, such as supervised desmopressin response testing. Source photo slide number 54 belongs to the classroom slide deck, not this 47-page PDF.

30 / SIADH: source causes and severity table

Lecture PDF p. 44



Original lecture image, including visible teaching annotations (PDF p. 44).

Source cause	Interpretation
Posterior-pituitary disease	Source describes slow constant VP leakage; inappropriate antidiuresis can occur with hypothalamic / pituitary or CNS disease, depending on the clinical setting.
Lung tumours	Small-cell lung carcinoma is the named example; ectopic production may be of actual AVP, rather than necessarily a merely “similar substance”.
Antidepressant medication history	Some drugs stimulate / potentiate inappropriate antidiuresis.

Source symptom grouping	Every listed feature
Mild to moderate	Weight gain, elevated BP, headache, nausea, vomiting, muscle weakness.
Advanced	Cerebral oedema → confusion, hallucination.
Grave	Seizures, coma, death.

Mechanism and qualification

VP remains inappropriately active for the osmotic state → kidney conserves too much water → plasma sodium diluted → extracellular hypotonicity → brain-cell water entry.

The source says “sodium deficit”, but dilutional hyponatraemia can reflect excess water relative to sodium without primary total-body sodium depletion.

Weight gain / elevated BP are retained from the source; typical established SIADH is often clinically euvolaemic, with no gross oedema and without obligatory hypertension. Severity depends on both the level and SPEED of sodium fall, not a rigid symptom sequence. Low sodium concentration and low total-body sodium content are different variables.

31 / Classroom case: Lucas and excess water

Lecture PDF pp. 23–25, 36–37, 44 | Original Photo 6 integrated with fluid regulation, not the anatomy PDF

Original case — retain the clinical details | Transcribed from classroom photo

Patient: Lucas, a 20-year-old Art student and experienced marathon runner.

- Roommate finds him disoriented and brings him to a nearby A&E.
- He has just completed a half-marathon in extreme heat (34°C), from Tsing Ma Bridge to Victoria Park.
- To avoid dehydration: copious water before the race, plus about another 3 L of bottled water during it.
- Back at the Student Hall: nausea and confusion.
- Assuming heat stroke, he continues drinking lots of cold bottled water.

Question: what could be his problem?

Answer approach:

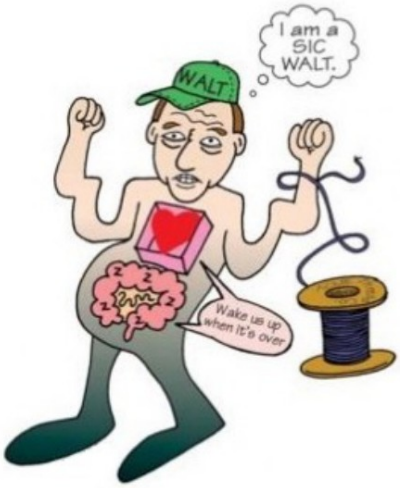
- The photographed answer is dilutional hyponatraemia. This is the leading intended explanation, not a sodium measurement supplied by the case.
- Large water intake over a short time without a matching electrolyte intake → water becomes excessive relative to exchangeable sodium / potassium.
- Sodium lost in sweat can contribute to the imbalance.
- [Extension] Exercise, stress and nausea can maintain non-osmotic VP secretion, limiting water excretion even as plasma sodium falls.
- Excess retained hypotonic fluid → extracellular osmolality ↓ → water enters cells; brain-cell swelling can explain nausea, confusion and disorientation.

Mechanistic clue	Answer reasoning beside the relevant concept
Water before / about 3 L during / more afterwards	Cumulative hypotonic intake relative to losses and excretory capacity matters; 3 L is not a universal toxicity threshold.
Heat / sweating	Sweat sodium loss can contribute. Because sweat is usually hypotonic, sweating alone does not establish dilutional hyponatraemia.
Exercise / nausea / stress	Non-osmotic VP stimuli can limit water excretion despite falling osmolality.
Confusion / disorientation	Compatible with hypotonic brain swelling; sodium measurement is needed to confirm the intended diagnosis and distinguish other exertional illness.

This is exercise-associated dilutional hyponatraemia in the photographed answer; it should not automatically be labelled tumour-related SIADH. The original clinical details are preserved. Compare Activity 6: DI plus restricted water produces water deficit / high sodium; Lucas's intended case produces relative water excess / low sodium.

32 / Hyperaldosteronism: causes, ions and symptoms

Lecture PDF p. 45



Original lecture image, including visible teaching annotations (PDF p. 45).

Written point	Complete source content
Definition	Excess aldosterone secretion.
Primary cause	Aldosterone-producing adenoma: Conn's syndrome.
Secondary cause	Abnormal RAAS activation, e.g. renovascular disease.
Major manifestations	Hypertension and hypokalaemia.

Source mnemonic	All eight features
A – S	Alkalosis; shallow respiration.
I – C	Irritability; constipation due to reduced gut motility.
W – A	Muscle weakness; arrhythmia / irregular heart rate.
L – T	Lethargy / fatigue; thready / weak pulse.

Correct the source's hypertension mechanism

Source: "hypertension due to hypernatremia".

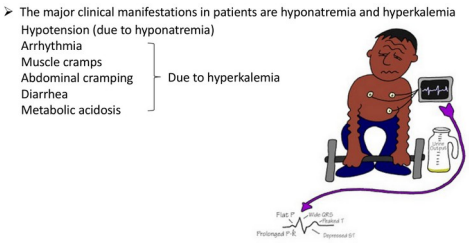
Better explanation: aldosterone-driven Na^+ retention expands ECF / effective circulating volume → BP rises. Plasma sodium may stay near normal because water regulation and pressure natriuresis limit the concentration change.

Distal K^+ wasting explains hypokalaemia; increased distal H^+ secretion contributes to metabolic alkalosis.

[Extension] Primary autonomous aldosterone excess usually suppresses renin; secondary renovascular RAAS activation raises renin. The mnemonic is a teaching list of possible manifestations, not a requirement that every patient has all eight signs.

33 / Hypoaldosteronism: causes and consequences

Lecture PDF p. 46



Original lecture image, including visible teaching annotations (PDF p. 46).

Written cause / outcome	Retained detail and qualification
Definition	Deficient aldosterone secretion / activity.
Adrenal insufficiency	Reduced adrenal hormone production.
Renal insufficiency	Source: chronic renal failure → low renin. This is relevant to hyporeninaemic hypoaldosteronism; not every chronic renal disease has low renin.
Drug causes	ACE inhibitors, ANGII-receptor blockers, renin inhibitors and related antihypertensive effects reducing aldosterone drive.
Major ion changes	Hyponatraemia and hyperkalaemia in the source.
Listed clinical manifestations	Hypotension; arrhythmia; muscle cramps; abdominal cramping; diarrhoea; metabolic acidosis.

Separate volume, potassium and acid–base effects

Na^+ loss / reduced ECF volume can lower BP; “hypotension due to hyponatraemia” is an oversimplification.
Reduced distal K^+ secretion → hyperkalaemia → cardiac / neuromuscular effects.
Reduced distal acid secretion and hyperkalaemia-related impaired ammonium handling contribute to metabolic acidosis; not every item is caused only by serum K^+.

The source labels its symptom bracket “due to hyperkalaemia”; the mechanistic grouping above avoids assigning low BP or the entire acid–base disturbance to potassium alone. “ACE blockers” in the source is usually termed ACE inhibitors. Aldosterone changes act through ENaC / Na^+–K^+-ATPase / ROMK-related transport, not by replacing the VP–AQP2 pathway.

34 / Electrolyte manifestations and final learning check

Lecture PDF p. 47 | One additional self-written integrated case

Imbalance	Main physiological consequence / manifestations
Hypernatraemia	Water deficit relative to sodium → extracellular hypertonicity / cell shrinkage; thirst, weakness and neurological dysfunction. Severe / rapid changes can cause confusion, seizures or coma.
Hyponatraemia	When hypotonic, water enters cells → headache, nausea, confusion; severe / rapid falls can cause cerebral oedema, seizures or coma.
Hyperkalaemia	Changes membrane excitability / conduction; muscle weakness or cramps and potentially dangerous arrhythmias. Hypoaldosteronism may also involve metabolic acidosis.
Hypokalaemia	Muscle weakness, reduced gut motility / constipation, lethargy, arrhythmias; severe cases can impair respiratory muscle function. Aldosterone excess can also cause metabolic alkalosis.

Integrated practice 2 | Self-written; not an original exam question

With preserved renal perfusion and a responsive renin system, an ACE inhibitor causes reduced ANGII / aldosterone action. Predict renin feedback, K^+ excretion and whether the defect is equivalent to a V_2-receptor mutation.

Answer approach:

- ANGII formation / signalling falls; loss of ANGII feedback tends to raise renin.
- Aldosterone drive falls → distal Na^+ reabsorption and K^+ secretion decrease → hyperkalaemia becomes more likely.
- This is not the same as renal VP resistance: V_2 / AQP2 responsiveness can remain intact.
- Urine volume / concentration still depend on VP, water intake, medullary gradient and filtration; do not infer a unique DI-like urine from an aldosterone effect alone.

All final learning objectives, retained

- State body-water distribution and identify major intracellular / extracellular electrolytes.
- Explain osmotic and haemodynamic sensing and BOTH physiological and behavioural responses.
- Explain how loop of Henle, collecting duct and VP coordinate urine concentration.
- List all FOUR ANGII functions and trace their target cells / circuits.
- Explain VP / aldosterone disorders and the clinical manifestations of all four electrolyte imbalances above.

This table synthesises the stated final objectives; supporting manifestations beyond the exact symptom mnemonics are explanatory extensions. Concentration, total-body content and extracellular volume must remain separate when interpreting sodium disorders.

35 / Coverage audit: nephron, gradient and body fluids

Lecture PDF pp. 1–19

PDF page	Key written content / source figures retained	Notes page
1	Correct title, lecturer and scope: countercurrent multiplier and urine formation.	1
2–3	Million nephrons, corpuscle/tubule, two capillary beds; cortical versus juxtamedullary locations/loops/efferent vessels; all diagram labels.	2
4	Passive opposite-stream exchange; co-current equilibration; more countercurrent transfer; original numerical examples.	3
5–6	Energy, descending/ascending permeability, ATPase, 3Na/2K, NKCC2 and bumetanide, Cl channel, repeated cycle; long-loop juxtamedullary emphasis.	4
7–8	Concentrates then dilutes; five figure steps; 300/100/1200 and axial gradient; longer loops; collecting-duct water arrows.	5
9	Vasa recta preserve, do not generate gradient; water/solute directions; <5% medullary flow; all labels.	6
10–11	Overhydration/VP↓/dilute urine; apical AQP2 versus basolateral AQP3/4; concentration needs permeability.	7
12–13	Dehydration→VP→principal-cell AQP2; protein-derived urea; tubular water loss concentrates urea; IMCD UT-A1/3 passive diffusion; interstitial osmotic force.	8
14	UT-A1 phosphorylation/apical accumulation; V2/Gs/AC/cAMP/PKA; UT-A3 exit; VP-dependent urea concentration.	9
15	Corticomedullary-to-papillary gradient; outer NaCl; inner NaCl+urea; nephron/vascular interaction.	10
16–17	Body-fluid section; TBW60%, ICF2/3 or40%BW, ECF1/3 or20%BW, plasma1/5ECF/interstitium4/5; 3/8/10% loss statements qualified.	11
18	Plasma/interstitial/intracellular ion chart, proteins/phosphate/Na/K differences and chart units.	12
19	Plasma→interstitium→ICF buffering; stable intracellular environment; physiological/behavioural response centres and hormones.	13

36 / Coverage audit: VP sensing and ANGII actions

Lecture PDF pp. 20–39

PDF page	Key written content / source figures retained	Notes page
20–21	PVN/SON magnocellular synthesis, axonal transport, posterior-pituitary terminals, vesicles and Herring-body qualification.	14
22	Stimulus/depolarisation/AP propagation, terminal Ca entry, vesicle exocytosis, fenestrated-capillary release.	15
23–24	1% sensitivity, water cartoon qualification, corrected osmolality definition, TRPV/pore/cation model, OVLT/SFO/PVN/SON, CVO barrier and 3V/CSF.	12,16
25	Cell shrink/swell, water out/in, cation conductance, depolarisation/hyperpolarisation, firing directions and figure voltages.	17
26–27	Exponential haemodynamic relationship, >~8% reduction, left atrial/arterial stretch, CNIX/X→NTS→CVLM/RVLM and direct sympathetic/hypothalamic outputs.	18
28–29	All excitatory/inhibitory circuit signs; normal/hypertension versus unloading; “no VP” qualification.	19–20
30	Osmolarity>280, atrial-volume/arterial-pressure input convergence on VP neurons.	28
31–32	V2, AQP2 Ser256, trafficking/actin depolymerisation; antidiuresis; 280/290mOsm,5pM,renal plateau/thirst/V1a graph qualified.	21
33	V1a systemic/coronary Ca/MLCK constriction; lecture macula-densa NO/PGE2→afferent JG renin route and feedback limitation.	22
34	Angiotensinogen/renin/ANGI/ACE/ANGII; all four output branches.	23
35	Adrenal cortical AT1→aldosterone→distal MR/HRE; basolateral ATPase, apical ENaC/ROMK; Na/water retention and K excretion.	24
36–37	Thirst definition, lamina terminalis=SFO/MnPO/OVLT; AT1/osmotic integration; cortical drinking and magnocellular PVN/SON VP routes.	25
38–39	Direct AT1/Ca/MLCK and central SFO/OVLT→MnPO→parvocellularPVN→RVLM; M/P distinctions and integrated schematic/micrograph.	26

37 / Coverage audit: disorders and classroom questions

Lecture PDF pp. 40–47 | The classroom photographs remain those already supplied

PDF page	All key content accounted for	Notes page
40–42	All image-only summaries: brain nuclei, posterior pituitary, dehydration, renin, ANGII, adrenal/brain/vessel/kidney branches; afferent stretch/NaCl/sympathetic triggers; final volume/resistance/BP outcomes.	27–28
43	CDI/NDI definitions; V2R/AQP2 membrane defect; X-linkedXq28/autosomal12q13; compensatory thirst and conditional hypernatraemia.	29
44	SIADH/posterior-pituitary disease/lung tumour/antidepressants; mild/advanced/grave symptom table; dilutional water excess and clinical qualifications.	30
45	Adenoma/Conn, renovascular activation, BP/K changes; all eight mnemonic features; hypertension mechanism corrected.	32
46	Adrenal/renal/drug causes, low-renin limitation, hypona/hyperK, all symptoms, acidosis and BP mechanisms qualified.	33
47	All final objectives and four electrolyte-imbalance manifestations.	34

Classroom item	Original details / answer reasoning retained	Notes page
Photos1–2	Baroreflex→VP, blood volume, vasodilation, cardiac sympathetic/inotropy and parasympathetic heart-rate/funny-current/conduction effects; NTS/CVC and zero-VP limitations.	19
Activity3	Small hypertonic saline; hyperosmolality, limited volume assumption, PVN/SON, posterior pituitary, apical AQP2, inner-medullary UT-A1, urea and concentrated small urine.	9
Activity4+vagotomy	Photographed aortic-dominant answer retained; competing carotid unloading, CNIX/X distinction and experimental assumptions explicit.	20
Activity6	48-year-old woman, no glycosuria, restricted water, lethargy/confusion, Na160/reference135–145, urine500/source>850, all five options, D best with subtype limitation.	29
Lucas	20-year-old Art student/experienced runner; half-marathon34°C/route; copious pre-race water/another3L/more after; all symptoms, intended dilutional diagnosis and sweating/VP qualifications.	31

Sources: Dr J Chu, BBMS2006 Counter-current Multiplier & Urine Formation / Body Fluid Homeostasis & Related Disorders, supplied student-version PDF (47 pages), plus the six previously supplied classroom photos. This corrected set replaces the anatomy-based study set for the present task. Original source wording is preserved when correction is needed; the explanation makes its limits visible. The worked questions are study interpretations, not an official key. Decorative section / cover pages carry no omitted teaching mechanisms.